JAVA-FSE-Complete

MODULE 15
Cloud Fundamentals

Aadi4-gif

Executive Summary

Cloud computing has transformed how organizations build, deploy, and scale applications by replacing traditional capital-intensive data centers with flexible, on-demand resources. This module provides a comprehensive introduction to cloud computing fundamentals with a strong focus on Amazon Web Services (AWS). It covers core cloud concepts, service and deployment models, compute, storage, networking, databases, and modern serverless architectures.

Introduction

- Traditional IT vs. Cloud Computing: Moving away from managing physical hardware and data centers to utilizing virtualized, on-demand resources over the internet.
- Virtualization & SOA: Virtualization abstracts physical hardware into virtual machines, while Service-Oriented Architecture (SOA) structures software as reusable services.
- Cloud vs. On-Premises: Evaluating cost models (CapEx vs. OpEx), scalability, maintenance overhead, and security.

Service Models:

- IaaS (Infrastructure as Service): Fundamental building blocks like virtual servers and storage (e.g., AWS EC2).
- PaaS (Platform as Service): Managed environments that let developers focus on app building without managing underlying infrastructure.
- SaaS (Software as Service): Fully functional, ready-to-use software applications delivered over the web.
- Deployment Models: Public, Private, Hybrid, and Community clouds.
- Major Providers: Overview of leading cloud providers including AWS, Microsoft Azure, and Google Cloud Platform (GCP).

Amazon EC2 & Container Basics

- Amazon EC2: Virtual servers in the cloud (instances) running on customizable AMIs (Amazon Machine Images). Managed using security groups (virtual firewalls) and key pairs for secure SSH access.

- Amazon ECS (Elastic Container Service): A fully managed container orchestration service that allows running and scaling Docker containers on AWS, bridging traditional virtual servers with modern containerized workloads.

AWS Storage Solutions: S3 & EBS

- Amazon S3 (Simple Storage Service): Object storage designed for scalability and durability. Utilizes buckets, objects, access permissions, diverse storage classes (Standard, Intelligent-Tiering, Standard-IA, Glacier), lifecycle policies, and versioning.
- Amazon EBS (Elastic Block Store): Block-level storage volumes designed for use with EC2 instances. Supports volume types (gp2, gp3, io1) and provides snapshots for automated backups.

Networking & Traffic Distribution: VPC & ELB

- Amazon VPC (Virtual Private Cloud): An isolated virtual network in the cloud containing public/private subnets, route tables, Internet Gateways (IGW), NAT Gateways, and VPC Peering to securely connect multiple VPCs.
- Elastic Load Balancer (ELB): Distributes incoming traffic across multiple targets.
- Application Load Balancer (ALB): Layer 7 load balancer optimized for HTTP/HTTPS traffic and path-based routing.
- Network Load Balancer (NLB): Layer 4 load balancer built for ultra-high performance and low latency handling of TCP/UDP traffic.

Serverless Computing: AWS Lambda & API Gateway

- AWS Lambda: An event-driven, serverless compute service that executes code in response to triggers (S3, API Gateway, DynamoDB Streams) on a pay-per-execution pricing model.

- AWS API Gateway: A fully managed service for creating, publishing, maintaining, and securing REST APIs at any scale, seamlessly integrating with serverless backends like Lambda across multiple deployment stages.

Conclusion

Mastering cloud fundamentals and AWS architecture enables engineers to design secure, highly available, and scalable cloud-based applications. From virtual servers and isolated networks to managed databases and serverless execution models, cloud platforms provide the comprehensive infrastructure needed for modern software development.